

Object Oriented Programming

(CS1004) CS All Sections

Course Instructor(s):

Mr. Shehreyar Rashid, Mr. Aqib Rehman, Ms. Munazza
Nida and Ms. Momina Behzad

Spring 2026

Sessional-I Exam

Total Time (Hrs):

1

Total Marks:

60

Total Questions:

1

Date: Feb 24, 2026

Do not write below this line.

Attempt all the questions.

[CLO1: Demonstrate the basic concepts of OOP]

Question 01 [60 marks]

What would be the output produced by executing the following C++ codes? Identify errors, if any (either write output or error (syntax/runtime), **both will not be accepted**). All the code snippet contains `#include<iostream>` and using namespace std; & these programs ^{are} compiled using g++ on a 32-bit system architecture.

```
char c[][6] = { "Exam", "Done", "OOPSI", "Crazy" };
char (*cp[][6]) = { c + 1, c + 3, c, c + 2 };
char (**cpp)[6] = cp + 1;
int main()
{
    cout << *((*cpp)[-1] + 4) << endl;
    cout << *(cpp[-1]) - 4 << endl;
    cout << **cpp << endl;
    (*(cp[3]))[5] = '!';
    cout << **((cpp + 2)) << endl;
    return 0;
}
```

Output/Error:

✓
Exam
Crazy
OOPSI. Crazy
43
four

```
struct IntArray{
    int * arr,size;
    void create(int *ptr,int s){
        size=s; arr=ptr;
    }
    void display(){
        for(int i=0;i<size;i++){
            cout<<arr[i]<<" ";
            cout<<endl;
        }
    };
    int main(){
        int arr[]={4,2,3,0,1};
        IntArray my_arr;
        my_arr.create(arr+2,3);
        arr[my_arr.arr[0]]=my_arr.arr[arr[1]];
        arr[arr[3]]=my_arr.arr[arr[1] + my_arr.arr[1] -1];
        my_arr.display();
        return 0;
    }
```

Output/Error:

4 1 3 1 1

National University of Computer and Emerging Sciences Islamabad Campus

```
void fun(int (*p)[2]) {
    cout << p[-1][-3] << " ";
    cout << p[-2][6] << "\n";
}

int main() {
    int arr[3][4] = {1,2,3,4,5,6,7,8,9,10,11,12};
    int (*ptr)[3] = (int(*)[3])(arr + 2) - 1;
    fun(((int(*)[2])++ptr));
    fun(ptr);
    return 0;
}
```

Output/Error:

~~Error~~
Line 9 → Syntax error
2.5

what is the error

```
struct Record {
    char *text;
};

void manipulate(Record r, char *p) {
    r.text = r.text + 1;
    *(r.text + 1) = *(p + 3);
    p = p + 2;
    *(p) = *(p - 2);
}

int main() {
    char arr[] = {'A','B','C','D','E','F','G','\0'};
    Record r1, r2;
    r1.text = arr;
    r2.text = r1.text + 3;
    char *p1 = r1.text + 2;
    char *p2 = r2.text - 1;
    *(p1) = *(p2 + 2);
    *(r2.text) = *(r1.text + 1);
    manipulate(r1, arr);
    r2.text = r1.text + 1;
    *(r2.text + 2) = '\0';
    *(r1.text + 5) = *(r1.text);
    cout << r1.text << endl;
    cout << r2.text << endl; return 0;
}
```

Output/Error:

A B A
B A

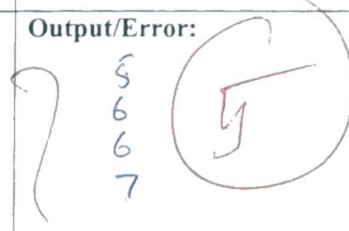
A B A
B A
S

```
struct Box {
    char *head, *mid, *tail;
} _b;

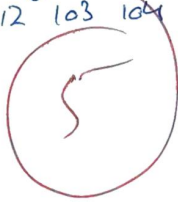
int main() {
    char *arr = "Exam01";
    _b.head = arr;    _b.mid = arr + 2;
    _b.tail = arr + 3;
    *_b.head = 'E';
    cout << "head: " << *_b.head << endl;
    cout << "mid: " << _b.mid << endl;
    cout << "tail: " << _b.tail << endl;
    return 0;
}
```

Output/Error:

Error
Line 8: Changing value of what?
4

<pre> struct Node { char *data; Node *next; }; void process(Node *p, int index) { if (p == nullptr) return; if (p->data[index] == '\0') { process(p->next, 0); return; } process(p, index + 1); if (p->next != nullptr) p->data[index] += 1; if (p->next != nullptr && index % 2 == 0) process(p->next, index); } void display(Node *p) { if (p == nullptr) return; cout << p->data << " "; display(p->next); } int main() { char a[] = "NNO"; char b[] = "AMAB"; char c[] = "DEMO"; Node n1, n2, n3; n1.data = a; n2.data = b; n3.data = c; n1.next = &n2; n2.next = &n3; n3.next = 0; process(&n1, 0); display(&n1); return 0; } </pre>	Output/Error: 
<pre> class integer{ int i; public: integer(int ii){i=ii;} int getI(){return i;} void setI(int ii){i=ii;} }; void display(integer i){ cout<<"Integer is "<<i.getI()<<endl;} void decrement(integer i){i.setI(i.getI()-1);} void increment_decrement(integer & i){ static int s; if(s == 0){ i.setI(i.getI()+1); s++; } } </pre>	Output/Error: 

National University of Computer and Emerging Sciences Islamabad Campus

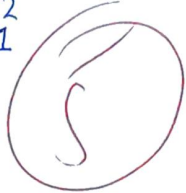
<pre> else{ decrement(i);s--; } display(i); } int main(){ integer i(5); display(i); increment_decrement(i); increment_decrement(i); increment_decrement(i); return 0; } </pre>	
<pre> struct Data { int* p; int size; }; void fun(int n, Data d, int index) { if (n <= 1 index >= d.size) return; d.p[index] += 10; cout << "n = " << n << " index = " << index << " value = " << d.p[index] << endl; fun(n - 2, d, index + 1); } int main() { int size = 5; int* arr = new int[size]; for (int i = 0; i < size; i++) arr[i] = 100 + i; Data d1; d1.p = arr; d1.size = size; cout << "Starting array values:" << endl; for (int i = 0; i < size; i++) cout << arr[i] << " "; cout << endl << endl; fun(6, d1, 0); cout << "\nFinal array values:" << endl; for (int i = 0; i < size; i++) cout << arr[i] << " "; cout << endl; delete[] arr; return 0; } </pre>	Output/Error: 100 101 102 103 104 n = 6 index = 0 value = 110 n = 4 index = 1 value = 111 n = 2 index = 2 value = 112 Final array values 110 111 112 103 104 

```
int** modifyPtr(int** &p) {
    return p;
}

int main() {
    int* x = new int; *x=42;
    int** y = &x;
    int** z = y;
    cout << **y << endl;
    delete *z;
    *modifyPtr(y) = nullptr;
    cout << (x == nullptr) << endl;
    return 0;
}
```

Output/Error:

42
1



```
struct Link {
    int id;
    Link* next;
};

struct Module {
    int id;
    Link* link;
};

int main() {
    Module A;
    A.id = 10;
    A.link = new Link;
    A.link->id = 20;
    A.link->next = new Link;
    A.link->next->id = 30;
    A.link->next->next = nullptr;
    cout << "STARTING TRACE" << endl;
    cout << "BEGIN-" << A.id << endl;
    Link* current = A.link;
    while(current != nullptr) {
        cout << " LINK-" << current->id << endl;
        current = current->next;
    }
    cout << "END-" << A.id << endl;
    current = A.link;
    while(current != nullptr) {
        Link* temp = current;
        current = current->next;
        delete temp;
    }
    return 0;
}
```

Output/Error:

STARTING TRACE
BEGIN-10
LINK-20
LINK-30
END-10



<pre>int main() { int a = 32, *ptr = &a; char ch = 'A', &cho = ch; cho += a; *ptr -= ch + 20; cout << a << " " << ch << endl; return 0; }</pre>	Output/Error: -85, A S
<pre>int main() { int x=8, *xx=&x, **xxx=&xx; cout<<sizeof(x)<<" "<<sizeof(xx)<<" "<<sizeof(xxx); return 0; }</pre>	Output/Error: 4, 4, 4 S